

MOSTAFA ALLAM

Software Engineer

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EXPERIENCE

Noon (Minutes Supply Chain — Order related)

Aug 2025 – Present

Software Development Engineer I (FastAPI, MySQL, GCP)

Hybrid - Cairo, Egypt

- Designed and launched the customer-facing Prints in Minutes service a document-to-print pipeline allowing users to upload PDFs and images and receive physical prints via Noon's logistics network.
- Built geospatial planning tools for network expansion, including automated serviceability polygons with metrics, direct polygon editing, ETA-based site ranking, and OSRM traffic intelligence.
- Integrated serialized inventory tracking for outbound orders, enabling item-level traceability and reducing fulfillment errors for high-value products.
- Developed the Driver Knowledge Hub a centralized training platform with SOPs cutting onboarding friction and reducing training-related support tickets.
- Revamped the cash-handling system with country-specific decimal precision, enabling expansion into new markets with different currency formats.
- Optimized core fulfillment services through query tuning, cache modernization, scalable workers, and data & index cleanup, reducing API latency by 50% and database load by 20%.

Darkmon

Feb 2025 – Jul 2025

Backend Engineer (Spring Boot, PostgreSQL)

Remote

- Built a TAXII 2 compliant server from scratch using Spring Boot 3, enabling standards-based cyber-threat intelligence exchange.
- Automated release processes, reducing deployment time and eliminating common manual-release errors.
- Eliminated N+1 queries across all system export workflows, substantially reducing database load and export latency.
- Collaborated with cross-functional teams to identify and resolve bugs and performance issues.

Cairo University - Faculty of Computers and Artificial Intelligence

Oct 2024 – Jun 2025

Part-time teaching assistant

Giza, Egypt

- Delivered tutorials in Operating Systems and Introduction to Software Engineering for approximately 280 students.
- Supported assignment discussions and evaluated student work, providing actionable technical feedback.

Agile Business Modules

Aug 2024 – Jan 2025

Full-stack Engineer (PHP, MySQL, VueJs, AngularJs)

Cairo, Egypt

- Worked on a new employees appraisals module in Modulus ERP system.
- Developed resource creation tool reducing time by 90%, boosting productivity.
- Refactored code base, reducing redundancy by approximately 30%, improving efficiency and maintainability.
- Revamped the vacation system to handle different vacation types and cycles.

EDUCATION

Cairo University - Faculty of Computers and Artificial Intelligence

Oct 2020 – Jul 2024

Bachelor of Software Engineering (3.73/4.00 GPA). Graduated with excellent distinction and honors.

Giza, Egypt

PROJECTS

Project Scaffolder (Graduation project) | Flask, ExpressJS, Jinja2, Docker, HTML

- Led the creation of a project scaffolding tool, enhancing development workflow automation for software teams.
- Collaborated on feature development and documentation to ensure maintainability and usability of the tool.

Courses Tracking Web-App | Java EE, Python, Golang, Sqlite, Docker, React

- Architected, developed and dockerized a microservices Course Tracking Web-App. With independent microservices for course management, data storage, and notifications, ensuring scalability, resilience and maintainability.
- Utilized JWT tokens for access control and authentication, Input validation, password hashing and data encryption.

Clinic Reservation Web-App | Golang, MySQL, RabbitMQ, Docker, OpenShift, React

- Spearheaded the development of a clinic reservation system. And deployed on OpenShift K8s platform.
- Integrated RabbitMQ messaging broker to enable asynchronous communication between system components, facilitating real-time updates and notifications for appointment scheduling, cancellations and availability changes.

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, Golang, Rust, PHP, JavaScript.

Technologies/Frameworks: Spring Boot, JPA, Java EE, Flask, Jinja2, GitHub, Docker, GCP, OpenShift, RabbitMQ, Fiber, Vue.js, AngularJs, MySQL, PostgreSQL, MongoDB.